

Tyrell SDK Developer's Guide

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Preface

About This Document

This document is distributed with the Tyrell NEXUS-6 SDK and provides an overview of the SDK system architecture, as well as describes how to use the Tyrell NEXUS-6 SDK to create custom applications for portable devices.

This document is intended to be used with the *Tyrell SDK API Reference* and assumes the reader has read the *Getting Started with the Tyrell SDK* document and has experience developing custom applications for portable devices.

 **NOTE** This document is intended to be printed using double-sided printing. If you print this document using single-sided printing, some pages appear blank.

Organization

The following table describes the chapters and appendixes in this document.

Chapter/Appendix	Description
Chapter 1, “Introduction to the Tyrell System Architecture”	Provides an overview of the Tyrell SDK system architecture.
Chapter 2, “Project Management”	Provides detailed information about how to create custom projects.
Chapter 3, “User Interface Management”	Provides detailed information about managing the user interface.
Chapter 4, “Media Management”	Provides detailed information about managing content.
Chapter 5, “System Management”	Provides detailed information about managing applications.
Appendix A, “Common Procedures”	Describes procedures that are common to the Tyrell SDK.
Appendix B, “Development System Default Configurations”	Describes the default configurations for the Replicant Engineering Mainboard and the NEXUS-169 CPU Daughtercard.

Document Conventions

Typographic The following table describes the typographic conventions used in this document.

Convention	Description	Example
<i>ABCdef</i>	Identifies book titles, emphasized words or words that appear in the glossary, and command variables.	You <i>must</i> log in as root. C:\> <i>cd directory_name</i> <i>path_name</i>
ABCdef	Identifies graphical user interface items with which you interact.	Click the OK button.
<u>ADCdef</u>	Identifies a hyperlink or URL.	http://www.janaowens.com
ABCdef	Identifies commands, computer-generated output, API elements, and code samples.	Build successful
[]	Indicates optional parameters within a syntax description.	[ALL]
{ }	Indicates a choice between parameters in a syntax description.	{ALL NONE}
	Separates items in a list of choices; used with braces ({ }) in a syntax description.	{ALL NONE}
...	Indicates that you can repeat a series of items one or more times.	<i>path_name</i> ...

Notational The following table describes the notational conventions used in this document.

Type	Symbol	Description
Tip		Provides information that can help you better utilize the product.
NOTE		Provides information that emphasizes the main text.
IMPORTANT		Provides important information required for the successful operation of the product.

Additional Information

The following table lists the documentation associated with the Tyrell NEXUS-6 SDK. To download the most recent versions, visit the Tyrell Extranet.

Resource	Description
<i>Tyrell SDK Release Notes</i>	Provides information about the latest Tyrell NEXUS-6 SDK release, including structural and operational changes.
<i>Tyrell SDK Defect Report</i>	Provides a description of all resolved and known defects for the latest Tyrell NEXUS-6 SDK release.
<i>Getting Started with the Tyrell SDK</i>	Provides an introduction to the Tyrell NEXUS-6 SDK and describes the procedures required to begin using the Tyrell NEXUS-6 SDK.
<i>Tyrell SDK API Reference</i>	Provides detailed information about the APIs that are distributed with the Tyrell SDK; for more information, see “SDK API Reference” on page 30.
<i>Orion Example Application User’s Guide</i>	Provides detailed information about the Orion Example Application.
<i>Cinema Example Application API Reference</i>	Provides detailed information about the Orion Example Application API.
<i>NEXUS-6 Product Data Sheet</i>	Provides detailed information about the NEXUS-6.
<i>Tyrell NEXUS-6 Reference Design Schematics</i>	Provides design details for NEXUS-6-based players.

Accessing the Tyrell Extranet

Introduction The Tyrell Extranet provides access to the following product resources:

- ◆ Release notes and defect reports
- ◆ Datasheets and product change notices (PCNs)
- ◆ Product documentation and schematics
- ◆ Source code and upgrades
- ◆ Example applications and application notes
- ◆ Return materials authorization (RMA) and host customization forms

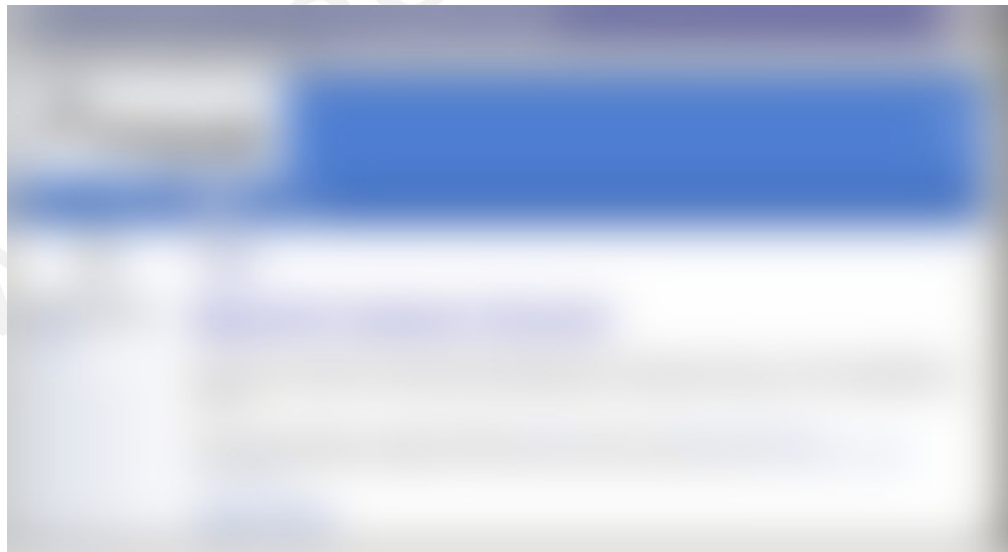
✍ **NOTE** To gain access to the Tyrell Extranet, you must purchase the Tyrell SDK and sign a license agreement. After signing the license agreement, a user name and password will be emailed to you.

Procedure To access the Tyrell Extranet:

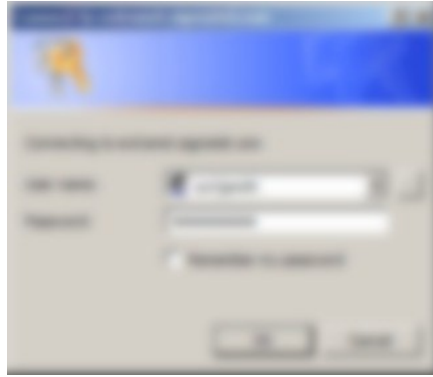
Step 1 Open a browser, and enter the following URL: <http://www.janaowens.com>

Step 2 From the links located on the left side of the page, click **Login**.

The Tyrell Customer Extranet Login page appears.



Step 3 Click **Login Now**.



Step 4 Enter your user name and password, and click **OK**.

 **NOTE** If you are unsure of your user name and password, contact your Tyrell representative.

Step 5 Once the Tyrell Extranet page appears, click **Products**, and then click the **NEXUS-6 Development** link.

Customer Support

Introduction You can request new features, report product defects, or ask general questions by visiting the following URL: <https://www.janaowens.com>

Request New Features To request new features, include the following information in your email:

- ◆ A detailed description of the requested feature, including a description of the user interface
- ◆ Feature specifications
- ◆ Project schedule
- ◆ Datasheet for peripherals

 **NOTE** Tyrell uses this information to prioritize new features.

Report a Defect To report a defect, include the following information in your email:

- ◆ *Tyrell NEXUS-6 SDK version number and part numbers*
Include the version number of your Tyrell NEXUS-6 SDK, and indicate whether any patches have been applied.
- ◆ *CPU product number*
Include the product number listed on the Tyrell CPU chip.
- ◆ *Full description of issue*

Include a detailed description of the issue, as well as answers to the following questions:

- Is the defect included in the *Tyrell SDK Defect Report*?
- Can you reproduce the defect with the latest SDK?
- What steps are required to reproduce the defect?

◆ *Debugging information* (if applicable)

Before contacting Tyrell Customer Support, you should attempt to debug the issue. If you are unable to resolve the issue, please include all relevant debugging information in your email submission.

◆ *Screenshots* (if applicable)

 **NOTE** When reporting multiple issues, submit only one issue per email.

CHAPTER 1

Introduction to the Tyrell System Architecture

Introduction This chapter describes the Tyrell software development kit (SDK) system architecture design and application programming interfaces (APIs).

In this Chapter The following table lists the sections in this chapter.

Section	Topic	Page
A	System Design	16
B	Application Framework	20
C	Application Programming Interfaces	25

Section A System Design

Introduction The Tyrell system architecture uses a top-down design that allows access to the API systems necessary to create custom applications. Developers can easily replace existing code or integrate custom code into this design.

This section describes the Tyrell SDK system design, including system organization, directory structure, source code, and project files.

In this Section The following table lists the topics in this section.

Topic	Page
Organization	17
Source Code	18
Project Files	19

Organization

Introduction Table 1-1 lists and describes the API systems into which the SDK system architecture is organized.

Table 1-1: Tyrell SDK API Systems

API System	Description
Application APIs	Provides the framework to build custom applications.
Component APIs	Provides access to the high-level APIs that allow developers to customize the SDK. For more information, see “Component APIs” on page 26.
Operating System API	Manages functionality at the operating system level. For more information, see “Operating System API” on page 27.
Device Driver API	Manages functionality at the device driver level. For more information, see “Device Driver API” on page 28.
Hardware API	Manages functionality at the hardware level. For more information, see “Hardware API” on page 29.

Directory Structure *Proprietary content removed.*

Source Code

Introduction	The Tyrell SDK system source code is required to develop advanced custom applications. This solution provides more flexibility and control over your custom applications.
Programming Languages	To enhance performance and maintain reliability, the core implementation of the Tyrell SDK is written in C and Assembly programming languages; however, some top-level Component APIs are written in C++.
File Types	<i>Proprietary content removed.</i>
Naming Convention	<i>Proprietary content removed.</i>

Project Files

Introduction Tyrell SDK source code is organized and managed using Green Hills project files (.gpj). These project files are compiled using the Green Hills Software MULTI Project Builder, which maintains file dependencies and sets options used for building the project.

Tyrell Naming Convention *Proprietary content removed.*

Section B Application Framework

Introduction The Application Framework provides the foundation to build custom unit tests or PMP applications. All source code, templates and other files are contained within the *application* directory.

Proprietary content removed.

In this Section The following table lists the topics in this section.

Topic	Page
Adding a custom board support package to your project	21
Memory Models	22
SDK Framework Source	23
SDK OS Media Player	24

Adding a custom board support package to your project

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

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Memory Models

Proprietary content removed.

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SDK Framework Source

Introduction *Proprietary content removed.*

No OS *Proprietary content removed.*

Basic OS *Proprietary content removed.*

SDK OS *Proprietary content removed.*

**SDK OS
Media** *Proprietary content removed.*

SDK OS Media Player

Introduction *Proprietary content removed.*

Firmware Updater *Proprietary content removed.*

Hostlink *Proprietary content removed.*

Player *Proprietary content removed.*

Section C Application Programming Interfaces

Introduction *Proprietary content removed.*

In this Section The following table lists the topics in this section.

Topic	Page
Component APIs	26
Operating System API	27
Device Driver API	28
Hardware API	29
SDK API Reference	30

Component APIs

Introduction *Proprietary content removed.*

Systems *Proprietary content removed.*

Operating System API

Introduction *Proprietary content removed.*

Subsystems *Proprietary content removed.*

Device Driver API

Introduction *Proprietary content removed.*

Subsystems *Proprietary content removed.*

Hardware API

Introduction *Proprietary content removed.*

Subsystems *Proprietary content removed.*

SDK API Reference

Introduction The *Tyrell SDK API Reference* provides detailed information about the APIs that are distributed with the Tyrell SDK. This section provides information about the location, organization, and navigation of the documentation.

Location The *Tyrell SDK API Reference* is available for download as an archived (.zip) file from the Tyrell Extranet. To download the file, complete these steps:

Step 1 Access the Tyrell Extranet as described in “Accessing the Tyrell Extranet” on page 12.

Step 2 From the NEXUS7XX Development page, click the **SDK** link.

Step 3 Locate the heading for your SDK product, and click the Tyrell **API Reference** link.

Step 4 Click **Save**, and save the file to a location on your local computer.

Step 5 Navigate to the directory in which you saved the file, and extract the file using an archive utility.

A compiled HTML help (.chm) file is extracted.

Step 6 Double-click the help file to open the *Tyrell SDK API Reference*.

Organization The information within the *Tyrell SDK API Reference* is organized into modules based on the structure of the Tyrell API systems. The top-level API modules are as follows:

- ◆ Component APIs
- ◆ Device Driver APIs
- ◆ Operating System APIs
- ◆ Hardware APIs

Detailed information about individual API systems are organized within each of these modules. For more information about the structure of the API systems, refer to “Application Programming Interfaces” on page 25.

Navigation You can locate detailed information about the API systems using either the HTML help search feature or the navigation pane.

CHAPTER 2

Project Management

Introduction This chapter provides the information necessary to create, manage, and debug your custom projects using the Tyrell SDK.

In this Chapter The following table lists the sections in this chapter.

Section	Topic	Page
A	Creating and managing custom projects	32
B	Debugging your custom projects	41
C	Using the UI Development Tool	52

See Also

- ◆ For detailed information about the APIs described in this chapter, refer to the *Tyrell SDK API Reference*.
- ◆ For information about the *Tyrell SDK API Reference*, see “SDK API Reference” on page 30.

Section A Creating and managing custom projects

Introduction This section describes the procedures necessary to create and manage your custom application projects.

In this Section The following table lists the topics in this section.

Topic	Page
Creating custom projects	33
Reconfigure an existing project	34
Adding additional subprojects	35
Configuration Files	36
Building custom projects	37
Changing the firmware version number	38
Requesting a customized host software package	39
Customizing vendor strings and product IDs	40

Creating custom projects

Introduction *Proprietary content removed.*

Before You Begin *Proprietary content removed.*

Syntax *Proprietary content removed.*

Arguments *Proprietary content removed.*

Configuration Files *Proprietary content removed.*

Naming Conventions *Proprietary content removed.*

Procedure *Proprietary content removed.*

Result *Proprietary content removed.*

Reconfigure an existing project

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

Adding additional subprojects

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

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Configuration Files

Introduction The configuration file is used to set all the options for your project(s).

Usage *Proprietary content removed.*

**Available
Configurations** *Proprietary content removed.*

Building custom projects

Introduction After you create your custom project, you can build it using the Tyrell command-line build script or using the Green Hills MULTI IDE.

Binary Output *Proprietary content removed.*

MULTI IDE *Proprietary content removed.*

Changing the firmware version number

Introduction *Proprietary content removed.*

Syntax *Proprietary content removed.*

Requesting a customized host software package

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

Customizing vendor strings and product IDs

Introduction This procedure describes how to customize vendor strings and product IDs for your custom player.

Procedure *Proprietary content removed.*

Section B Debugging your custom projects

Introduction Debugging the Tyrell SDK consists of two components, the GHS slingshot and MULTI IDE application and the Remote Procedural Call (RPC) instrumentation subsystem. The RPC instrumentation subsystem allows you to monitor and control your custom applications using the UART connection on the NEXUS7XX Development System and RPCNET, a python application that runs on your computer.

Before You Begin Before you begin, verify that the SlingShot debug device is installed and connected to a target. For more information about connecting to a target, refer to the Green Hills project documentation.

In this Section The following table lists the topics in this section.

Topic	Page
Connecting and configuring the NEXUS-6 Development System	42
Running the RPCNet application	43
Adding TSS LogText to your project	44
Debugging aborts	45
Debugging SDRAM Applications with the Green Hills MULTI IDE	46
Debugging the SDRAMLess player	47
MULTI Debugger Register View Window	48
Browsing the function call tree in MULTI Debugger	49
Browsing ThreadX information in the MULTI Debugger	50
Troubleshooting common MULTI Debugger issues	51

See Also

- ◆ *Getting Started with the Tyrell SDK*
- ◆ Green Hills Software product documentation

Connecting and configuring the NEXUS-6 Development System

Introduction *Proprietary content removed.*

Configuration *Proprietary content removed.*

Procedure *Proprietary content removed.*

Running the RPCNet application

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

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Adding TSS LogText to your project

Introduction *Proprietary content removed.*

Source Code *Proprietary content removed.*

Procedure *Proprietary content removed.*

Result *Proprietary content removed.*

Debugging aborts

Introduction *Proprietary content removed.*

**How do I
Know an
Abort has
Occurred?** *Proprietary content removed.*

**How do I
Debug an
Abort?** *Proprietary content removed.*

Debugging SDRAM Applications with the Green Hills MULTI IDE

Introduction *Proprietary content removed.*

Before You Begin Before you begin this procedure you must install the Slingshot debug device as described in *Getting Started with the Tyrell SDK*.

Procedure *Proprietary content removed.*

Debugging the SDRAMLess player

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

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MULTI Debugger Register View Window

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

Browsing the function call tree in MULTI Debugger

Introduction Function calls can be browsed in the Tree Browser Window of the MULTI Debugger.

MULTI Debugger Navigation Bar This image provides an example of the MULTI Debugger Navigation Bar.



Static Calls by Function To browse static calls by function:

Step 1 In the MULTI Debugger Navigation Bar, select the function whose static calls you want to browse.

Step 2 Select **Static Calls** from the Browse menu.
The Static Call Browser dialog box appears.

Static Calls by File To browse static calls by file:

Step 1 In MULTI Debugger Navigation Bar, select the file whose static calls you want to browse.

Step 2 Select **File Calls** from the Browse menu.
The Static Call Browser dialog box appears.

Dynamic Calls by Function To browse dynamic calls by function:

Step 1 In MULTI Debugger Navigation Bar, select the function whose dynamic calls you want to browse.

Step 2 Select Dynamic Calls from the Browse menu.
The Dynamic Call Browser dialog box appears.

 **NOTE** Before you can browse the dynamic call tree of a function, profiling information must be collected. For more information, refer to Chapter 15, “Using the Profiler” of the *MULTI:Debugging* document.

Browsing ThreadX information in the MULTI Debugger

Introduction One of most powerful features of the MULTI Debugger is that it allows you to view real-time information about ThreadX components, such as threads, semaphores, and mutexes. The following procedure describes how to use this feature.

Procedure To browse ThreadX component information:

Step 1 Verify that the program is not running in the debugger window.

If the program is running, click the Stop  button to stop the debugger.

Step 2 Click on the **ThreadX**  button.

The ThreadX Information dialog box appears.



Step 3 Click a ThreadX component to view its information.

For example, if you click the Threads button, a new dialog box appears that lists the threads, their priority, their state, and other related information.

Troubleshooting common MULTI Debugger issues

Introduction This section lists some issues that you might face with the MULTI debugger and how to troubleshoot them.

Problem *Proprietary content removed.*

Most Likely Cause *Proprietary content removed.*

Solution *Proprietary content removed.*

Section C Using the UI Development Tool

Introduction The User Interface Development Tool simulates the NEXUS-6 Development System on a Win32 platform. Because the User Interface Development Tool does not require the NEXUS-6 Development System, developers can work together to develop user interface firmware without requiring individual development systems.

Required Tools *Proprietary content removed.*

In this Section The following table lists the topics in this section.

Topic	Page
Building and debugging User Interface Development Tool projects	53
User Interface Development Tool Player application	54
User Interface Development Tool HostLink application	55
Customizing the User Interface Development Tool Keypad buttons	56
Changing the LCD resolution	57
Testing the player interface	58
Application Programming Interface	59

Building and debugging User Interface Development Tool projects

Introduction To use the User Interface Development Tool, you must build and debug the User Interface Development Tool project files.

Before You Begin Before you begin using the User Interface Development Tool, verify the required tools are installed and create your custom project as described in “Creating custom projects” on page 33.

Project Files *Proprietary content removed.*

Procedure *Proprietary content removed.*

User Interface Development Tool Player application

Introduction *Proprietary content removed.*

**Player
Dialog Box** *Proprietary content removed.*

**Player
Interface** *Proprietary content removed.*

**Keypad
Dialog Box** *Proprietary content removed.*

**UI Test
Dialog Box** *Proprietary content removed.*

User Interface Development Tool HostLink application

Introduction *Proprietary content removed.*

**HostLink
Dialog Box** *Proprietary content removed.*

Host Interface *Proprietary content removed.*

**USB Dialog
Box** *Proprietary content removed.*

Customizing the User Interface Development Tool Keypad buttons

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

Changing the LCD resolution

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

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Testing the player interface

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

Custom Tests *Proprietary content removed.*

Proprietary code removed.

Code Sample 2-1. Adding custom tests to the User Interface Development Tool

Application Programming Interface

Introduction *Proprietary content removed.*

Audio *Proprietary content removed.*

Display *Proprietary content removed.*

Graphics *Proprietary content removed.*

OS *Proprietary content removed.*

Queue *Proprietary content removed.*

Text *Proprietary content removed.*

Utilities *Proprietary content removed.*

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CHAPTER 3

User Interface Management

Introduction This chapter provides the information necessary to manage a portable device user interface using the Tyrell SDK.

In this Chapter The following table lists the sections in this chapter.

Section	Topic	Page
A	User Interface Manager	62
B	Text Support	72
C	Graphics Support	76
D	Device Driver Display Interface	84
E	Resource Management	91
F	Input Translation	98
G	User Settings Interface	103

See Also

- ◆ For detailed information about the APIs described in this chapter, refer to the *Tyrell SDK API Reference*.
- ◆ For information about the *Tyrell SDK API Reference*, see “SDK API Reference” on page 30.

Section A User Interface Manager

Introduction The User Interface Manager (UIM) API provides the functionality necessary to manage user interface elements. This section describes the UIM API.

In this Section The following table lists the topics in this section.

Topic	Page
Overview	63
UIM Widgets	64
Widget Tree	65
Widget Drawing	66
UIM System Messages	67
Creating a custom widget	68
Adding a user interface window	69
Creating a fly-out menu	70
Adding an animation sequence	71

Overview

Introduction *Proprietary content removed.*

UIM Widgets *Proprietary content removed.*

Widget Classes *Proprietary content removed.*

For more information about these class structures, refer to the Component APIs module of the *Tyrell SDK API Reference*.

Subclassing *Proprietary content removed.*

Widget Class Functions *Proprietary content removed.*

Widget Properties *Proprietary content removed.*

UIM Widgets

Animation *Proprietary content removed.*

Base *Proprietary content removed.*

Bitmap *Proprietary content removed.*

Number *Proprietary content removed.*

Popup *Proprietary content removed.*

Progress *Proprietary content removed.*

Scroll *Proprietary content removed.*

Slider *Proprietary content removed.*

Stringlist *Proprietary content removed.*

Text *Proprietary content removed.*

Time and Date *Proprietary content removed.*

See Also For more information about the widgets described in this section, refer to the *Tyrell SDK API Reference*.

Widget Tree

Introduction *Proprietary content removed.*

Guidelines *Proprietary content removed.*

Example *Proprietary content removed.*

Widget Drawing

Introduction *Proprietary content removed.*

**Active
Clipping
Region** *Proprietary content removed.*

Guidelines *Proprietary content removed.*

UIM System Messages

Introduction *Proprietary content removed.*

Message Handling *Proprietary content removed.*

Call Stack *Proprietary content removed.*

Modal Call Stack *Proprietary content removed.*

Focus Call Stack *Proprietary content removed.*

Creating a custom widget

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

Adding a user interface window

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

Creating a fly-out menu

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

Adding an animation sequence

Introduction *Proprietary content removed.*

See Also For more information about the functionality described in this procedure, see “UIM Widgets” on page 64 or refer to the User Interface Module of the *Tyrell SDK API Reference*.

Procedure *Proprietary content removed.*

Section B Text Support

Introduction Text support, such as encoding and font and language support, is provided by the Text API. This section describes the Text API system.

In this Section The following table lists the topics in this section.

Topic	Page
Text Encoding	73
Font Support	74
Language Support	75

Text Encoding

Introduction *Proprietary content removed.*

Supported Encodings *Proprietary content removed.*

UTF-16 *Proprietary content removed.*

UTF-8 *Proprietary content removed.*

DBCS *Proprietary content removed.*

ASCII *Proprietary content removed.*

Font Support

Introduction *Proprietary content removed.*

Initialization *Proprietary content removed.*

Font Structure *Proprietary content removed.*

Language Support

Introduction *Proprietary content removed.*

Code Pages *Proprietary content removed.*

Text Entry *Proprietary content removed.*

Sort Order *Proprietary content removed.*

Localization *Proprietary content removed.*

Section C Graphics Support

Introduction Graphics support, such as bitmap drawing and basic image manipulation, is provided by the Graphics API system. This section describes the Graphics API system.

In this Section The following table lists the topics in this section.

Topic	Page
Device Contexts	77
Bitmap Drawing	78
Basic Image Manipulation	79
Alpha Blending	80
Bitmap Resampling	81
Album Art	82
Adding support for a graphics format	83

Device Contexts

Introduction *Proprietary content removed.*

Implementation *Proprietary content removed.*

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Bitmap Drawing

Introduction *Proprietary content removed.*

Implementation *Proprietary content removed.*

**Memory
Structure** *Proprietary content removed.*

**1BPP
Horizontal** *Proprietary content removed.*

1BPP Vertical *Proprietary content removed.*

**16BPP–
RGB565** *Proprietary content removed.*

**16BPP–
RGB565Direct
Draw** *Proprietary content removed.*

**18BPP–
RGB666** *Proprietary content removed.*

Basic Image Manipulation

Introduction *Proprietary content removed.*

Text Strings *Proprietary content removed.*

Line Drawing *Proprietary content removed.*

Rectangles *Proprietary content removed.*

Alpha Blending

Introduction *Proprietary content removed.*

Transparency *Proprietary content removed.*

Transparency Levels *Proprietary content removed.*

Alpha Values *Proprietary content removed.*

Bitmap Drawing *Proprietary content removed.*

Chroma Color Keying *Proprietary content removed.*

Alpha Blending with Widgets *Proprietary content removed.*

Bitmap Resampling

Introduction *Proprietary content removed.*

**Resampling
Filters** *Proprietary content removed.*

**Resampling
Example** *Proprietary content removed.*

**Stretch Blitting
Concepts** *Proprietary content removed.*

**Stretch Blitting
Examples** *Proprietary content removed.*

**Non-Widget
Stretch Blitting** *Proprietary content removed.*

Album Art

Introduction *Proprietary content removed.*

**Supported
Types** *Proprietary content removed.*

**Album Art
APIs** *Proprietary content removed.*

Adding support for a graphics format

Introduction *Proprietary content removed.*

**Format
Structure** *Proprietary content removed.*

**Supported
Bitmaps** *Proprietary content removed.*

**Bitmap Copy
Acceleration** *Proprietary content removed.*

Section D Device Driver Display Interface

Introduction The Device Driver Display Interface provides the functionality necessary to send images to the display. This sections describes the Display Interface.

In this Section The following table lists the topics in this section.

Topic	Page
Overview	84
Controller Support Module	86
Video Synchronization	87
Motion Video Display Programming	88
Hardware Consideration for Video Application	89
Adding support for a display controller	90

Overview

Proprietary content removed.

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Controller Support Module

Introduction *Proprietary content removed.*

For more information, see “Adding support for a display controller” on page 90.

Sample Code *Proprietary content removed.*

Video Synchronization

Introduction *Proprietary content removed.*

VSYNC Mode *Proprietary content removed.*

**Boot time
reduction** *Proprietary content removed.*

Motion Video Display Programming

VSYNC Mode *Proprietary content removed.*

**LCD
Controller
Driver** *Proprietary content removed.*

**Video
Application** *Proprietary content removed.*

Hardware Consideration for Video Application

LCD Module *Proprietary content removed.*

SDRAM *Proprietary content removed.*

Adding support for a display controller

Introduction *Proprietary content removed.*

**Required
Functionality** *Proprietary content removed.*
For more information these functions, refer to the *Tyrell SDK API Reference*.

Commands *Proprietary content removed.*
For more information about these and other commands, refer to the *Tyrell SDK API Reference*.

Build Options *Proprietary content removed.*

Procedure *Proprietary content removed.*

Result *Proprietary content removed.*

Section E Resource Management

Introduction File system resources are managed using the Green Hills MULTI IDE and a custom file system implementation managed by the Resource Manager API and the File System API. This section describes how to manage resources using the MULTI IDE and the Resource Manager API.

In this Section The following table lists the topics in this section.

Topic	Page
Overview	92
MULTI Resource Bundle	93
Creating font resource projects	94
Building font resource projects	95
Adding resources and setting resource options	96
Creating a string resource	97

Overview

Introduction *Proprietary content removed.*

**Resource ID
Caching** *Proprietary content removed.*

**Compiler
Macros** *Proprietary content removed.*
For more information about these compiler macros, refer to the Resource
Manager module of the *Tyrell SDK API Reference*.

**Resource
Building** *Proprietary content removed.*

MULTI Resource Bundle

Introduction *Proprietary content removed.*

**Resource
Types** *Proprietary content removed.*

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Creating font resource projects

Introduction *Proprietary content removed.*

For more information about font resources, see Chapter 3, “Text Support.”

Location *Proprietary content removed.*

Syntax *Proprietary content removed.*

Supported Image Types *Proprietary content removed.*

Image Naming Convention *Proprietary content removed.*

Before You Begin *Proprietary content removed.*

Procedure *Proprietary content removed.*

Result *Proprietary content removed.*

Building font resource projects

Introduction *Proprietary content removed.*

**Resource
Object File** *Proprietary content removed.*

Procedure *Proprietary content removed.*

Adding resources and setting resource options

Introduction *Proprietary content removed.*

For more information about resources and resource management, see Chapter 3, “User Interface Management.”

Procedure *Proprietary content removed.*

Creating a string resource

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

Sample Code *Proprietary content removed.*

Section F Input Translation

Introduction *Proprietary content removed.*

In this Section The following table lists the topics in this section.

Topic	Page
Overview	99
Input Source and Mapping Tables	100
Button Driver Interface	101
Event Translation	102

Overview

Introduction *Proprietary content removed.*

Input Devices *Proprietary content removed.*

Buttons *Proprietary content removed.*

Input Source and Mapping Tables

Introduction *Proprietary content removed.*

Input Source Table *Proprietary content removed.*

Mapping Table *Proprietary content removed.*

Button Driver Interface

Introduction *Proprietary content removed.*

For more information about translating these event messages, see “Event Translation” on page 102.

Message Structure *Proprietary content removed.*

Examples *Proprietary content removed.*

Event Translation

Introduction *Proprietary content removed.*

Event Messages *Proprietary content removed.*

Single-Click *Proprietary content removed.*

Double-Click *Proprietary content removed.*

Down and Hold *Proprietary content removed.*

Section G User Settings Interface

Introduction *Proprietary content removed.*

Operation *Proprietary content removed.*

APIs *Proprietary content removed.*

For more information about the APIs listed in this section, refer to *Tyrell SDK API Reference*.

User Setting Structures *Proprietary content removed.*

Application Subgroups *Proprietary content removed.*

For more information about USI and Settings structures, refer to the *Tyrell SDK API Reference*.

Tags *Proprietary content removed.*

Application Versions *Proprietary content removed.*

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CHAPTER 4

Media Management

Introduction This chapter provides the information necessary to manage media on a portable device using the Tyrell SDK.

In this Chapter The following table lists the sections in this chapter.

Section	Topic	Page
A	Content Management Interface	106
B	Play Queue Interface	117
C	Media Engine Interface	120
D	Post Processing Manager	123
E	Image Processing Manager	128

See Also

- ♦ For detailed information about the APIs described in this chapter, refer to the *Tyrell SDK API Reference*.
- ♦ For information about the *Tyrell SDK API Reference*, see “SDK API Reference” on page 30.

Section A Content Management Interface

Introduction *Proprietary content removed.*

In this Section The following table lists the topics in this section.

Topic	Page
Overview	107
CMI Subcomponents	108
CMI Clients	109
Sorted Lists	110
Selection Lists	113

Overview

Proprietary content removed.

For more information about metadata sorted lists, see “Sorted Lists” on page 110.

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CMI Subcomponents

Introduction *Proprietary content removed.*

Content Manager *Proprietary content removed.*

Query Manager Interface *Proprietary content removed.*

Database Access Interface *Proprietary content removed.*

Sort Module Interface *Proprietary content removed.*

Playlist File Format Interface *Proprietary content removed.*

CMI Clients

Introduction *Proprietary content removed.*

User Interface Manager *Proprietary content removed.*

For more information about UIM, see Chapter 3, “User Interface Management.”.

Play Queue Manager *Proprietary content removed.*

Audio Stream and Filtergraph *Proprietary content removed.*

Sorted Lists

Introduction *Proprietary content removed.*

**Sorted List
Hierarchy and
Level-ness** *Proprietary content removed.*

**Folder Sorted
Lists** *Proprietary content removed.*

**Metadata
Sorted Lists** *Proprietary content removed.*

**User Ratings
Sorted List** *Proprietary content removed.*

**Recorded
Sorted Lists** *Proprietary content removed.*

**Delete Media
Handles** *Proprietary content removed.*

**RAM-Based
Memory Usage** *Proprietary content removed.*

Sample Code *Proprietary content removed.*

Adding a Custom Sorted List in File-Based CMI Solution

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

Adding a Custom Sorted List in RAM-Based CMI Solution

Proprietary content removed.

Selection Lists

Introduction *Proprietary content removed.*

Adding a Custom Selection List *Proprietary content removed.*

Appending an Element to a Selection List *Proprietary content removed.*

Removing an Element from a Selection List *Proprietary content removed.*

Removing all Elements from a Selection List *Proprietary content removed.*

Validating a Selection List (not currently supported) *Proprietary content removed.*

Album Art

Proprietary content removed.

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Configuration

Introduction *Proprietary content removed.*

**File-Based vs
RAM-Based** *Proprietary content removed.*

**Maximum
Number of
Files
Supported** *Proprietary content removed.*

**Sorted List
Switches** *Proprietary content removed.*

**Folder Sorting
Configuration** *Proprietary content removed.*

**Sorting Tracks
Under Album** *Proprietary content removed.*

**Partial String
Caching for
File-Based
CMI** *Proprietary content removed.*

**Metadata
Extraction
Deferral
Switch** *Proprietary content removed.*

**Metadata -
Disabling
Checking all
Codecs on
Filetype
Failure** *Proprietary content removed.*

**Album Art
Total Disable
in Now Playing
Window
Example** *Proprietary content removed.*

**Album Art
Usage Case
Switches** *Proprietary content removed.*

Event Handling

Introduction *Proprietary content removed.*

Player Startup *Proprietary content removed.*

Player Shutdown *Proprietary content removed.*

MMC Hot Insertion *Proprietary content removed.*

MMC Hot Ejection *Proprietary content removed.*

Section B Play Queue Interface

Introduction The Play Queue Interface (PQI) provides functionality to manage the playqueue.

In this Section The following table lists the topics in this section.

Topic	Page
Overview	118
Adding Media to the Playqueue Automatically	119

Overview

Introduction *Proprietary content removed.*

Play Queue Manager for Different SDK Versions *Proprietary content removed.*

Update a Play Queue with Sorted Lists *Proprietary content removed.*

Play Queue Types *Proprietary content removed.*

Playback Modes *Proprietary content removed.*
For more information about this function, refer to the *Tyrell SDK API Reference*.

Shuffle Play Queue *Proprietary content removed.*

PQM Interaction *Proprietary content removed.*

Adding Media to the Playqueue Automatically

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

Section C Media Engine Interface

Introduction The Media Engine API provides functionality to manage audio streams on a portable device. This section describes the functionality and capabilities of the Media Engine system.

In this Section The following table lists the topics in this section.

Topic	Page
Overview	121
Framework Media Engine Interface	122

Overview

Introduction *Proprietary content removed.*

Subsystems *Proprietary content removed.*

Configurations *Proprietary content removed.*

For a complete list of available configuration option, see “Creating and managing custom projects” on page 32.

Framework Media Engine Interface

Introduction *Proprietary content removed.*

**Messaging:
Media Engine
to UI** *Proprietary content removed.*

Capabilities *Proprietary content removed.*

Modes *Proprietary content removed.*

**Audio and
Video
Playback
Controls** *Proprietary content removed.*

**Recording
Controls** *Proprietary content removed.*

**FM Tuner
Controls** *Proprietary content removed.*

Section D Post Processing Manager

Introduction *Proprietary content removed.*

This section describes the PPM API system.

In this Section The following table lists the topics in this section.

Topic	Page
Overview	124
Mixer	125
Effects Manager	126
Audio Effects	127

Overview

Introduction *Proprietary content removed.*

Mix Points For more information, see “Mixer” on page 125.

**Effects
Processing** *Proprietary content removed.*

For more information, see “Audio Effects” on page 127.

Thread Safety *Proprietary content removed.*

Mixer

Introduction *Proprietary content removed.*

Buffered Stream *Proprietary content removed.*

Direct-Callback Stream *Proprietary content removed.*

Mixer Manager *Proprietary content removed.*

Thread Safety *Proprietary content removed.*

Effects Manager

Introduction *Proprietary content removed.*

**Volume
Adjustments** *Proprietary content removed.*

APIs *Proprietary content removed.*

Sample Code *Proprietary content removed.*

Proprietary code removed.

Code Sample 4-1. Controlling the GEQ

Audio Effects

Introduction *Proprietary content removed.*

Initialization *Proprietary content removed.*

Proprietary code removed.

Code Sample 4-2. Initializing an audio effect

Effect Enabling *Proprietary content removed.*

Proprietary code removed.

Code Sample 4-3. Enabling an audio effect

Property Management *Proprietary content removed.*

Proprietary code removed.

Code Sample 4-4. Setting audio effect properties

Effect Processing *Proprietary content removed.*

Proprietary code removed.

Code Sample 4-5. Processing an effects callback

Shutdown *Proprietary content removed.*

Proprietary code removed.

Code Sample 4-6. Shutting down an effect

Effects Unit Test *Proprietary content removed.*

Section E Image Processing Manager

Introduction *Proprietary content removed.*

Imagewidget *Proprietary content removed.*

Photoalbum *Proprietary content removed.*

**Regular vs
Direct Draw** *Proprietary content removed.*

CHAPTER 5

System Management

Introduction This chapter provides the information necessary to manage portable device systems using the Tyrell SDK.

In this Chapter The following table lists the sections in this chapter.

Section	Topic	Page
A	File System	130
B	System Threads and Interrupt Service Routines	131
C	System Messaging	143
D	Event Processing	150
E	RTC, Alarm, Timers, and Persistent Bits	159
F	Power Management	166
G	Memory Management	171
H	Headphone Short Detection	180

See Also

- ◆ For detailed information about the APIs described in this chapter, refer to the *Tyrell SDK API Reference*.
- ◆ For information about the *Tyrell SDK API Reference*, see “SDK API Reference” on page 30.

Section A File System

Introduction *Proprietary content removed.*

Memory Allocation *Proprietary content removed.*

File Access *Proprietary content removed.*

Proprietary code removed.

Code Sample 5-1. Specifying absolute file addresses

Error Codes *Proprietary content removed.*

Reentrancy *Proprietary content removed.*

For more information about Resource Management, see Chapter 3, “User Interface Management.”

Long File Names *Proprietary content removed.*

Proprietary code removed.

Code Sample 5-2. Converting a short file name to a long file name

Section B System Threads and Interrupt Service Routines

Introduction The following API subsystems provide functionality to manage the FAT32 file system, system threads, and interrupt service routines:

Proprietary content removed.

This section describes these API subsystems.

In this Section The following table lists the topics in this section.

Topic	Page
System Threads	132
Tyrell ThreadX Interface	133
Kernel Objects	134
Interrupt Service Routines	135
Creating system threads	136
Releasing and terminating system threads	137
Creating, sending, and releasing message queue objects	138
Creating and releasing semaphore objects	139
Creating and releasing mutex objects	140
Creating and releasing timer objects	141
Registering an interrupt service routine	142

System Threads

Introduction *Proprietary content removed.*

Player *Proprietary content removed.*

Hostlink *Proprietary content removed.*

Custom *Proprietary content removed.*

Tyrell ThreadX Interface

Introduction *Proprietary content removed.*

**Conversion
Macros** *Proprietary content removed.*

Proprietary code removed.

Code Sample 5-3. Putting a thread to sleep using a THI conversion macro

**Memory
Management** *Proprietary content removed.*

For more information about THI kernel objects, see “Kernel Objects” on page 134.

Kernel Objects

Introduction *Proprietary content removed.*

Semaphore *Proprietary content removed.*

For information about creating and releasing a semaphore object, see “Creating and releasing semaphore objects” on page 139.

Mutex *Proprietary content removed.*

For information about creating and releasing a mutex object, see “Creating and releasing mutex objects” on page 140.

Message Queue *Proprietary content removed.*

For information about creating and releasing a message queue object, see “Creating, sending, and releasing message queue objects” on page 138.

Timer *Proprietary content removed.*

For information about creating and releasing a timer object, see “Creating and releasing timer objects” on page 141.

Interrupt Service Routines

Introduction *Proprietary content removed.*

Interrupt Sources *Proprietary content removed.*

ICOLL *Proprietary content removed.*

DSR *Proprietary content removed.*

Handler Lookup Table *Proprietary content removed.*

Creating system threads

Introduction *Proprietary content removed.*

Proprietary code removed.

Code Sample 5-4. Creating system threads

Releasing and terminating system threads

Introduction *Proprietary content removed.*

Sample Code *Proprietary content removed.*

Proprietary code removed.

Code Sample 5-5. Releasing and terminating system threads

Creating, sending, and releasing message queue objects

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

Creating and releasing semaphore objects

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

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Creating and releasing mutex objects

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

Creating and releasing timer objects

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

See Also For more information, see “RTC, Alarm, Timers, and Persistent Bits” on page 159.

Registering an interrupt service routine

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

Sample Code *Proprietary content removed.*

Proprietary code removed.

Code Sample 5-6. Registering an interrupt service routine

Section C System Messaging

Introduction This section describes system messaging and the API that is provided by the Tyrell SDK to manage system messages.

In this Section The following table lists the topics in this section.

Topic	Page
Overview	144
Creating and deleting message queue objects	145
Adding a message group	146
Adding messages to a group	147
Receiving messages	148
Sample Code	149

Overview

Introduction *Proprietary content removed.*

For more information on how to create and release message queues, see “Creating and deleting message queue objects” on page 145.

Message Queue Data Structure *Proprietary content removed.*

Message Filtering *Proprietary content removed.*

Message Build Macros *Proprietary content removed.*

Creating and deleting message queue objects

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

See Also For an example of how to create and delete message queue objects, see Sample Code 5-7 on page 149.

For more information on ThreadX API functions, refer to the *Tyrell SDK API Reference*.

Adding a message group

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

Adding messages to a group

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

See Also For an example of how to define message in a group, see Sample Code 5-8 on page 149.

Receiving messages

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

See Also For an example of how to receive messages, see Sample Code 5-10 on page 149.

Sample Code

Sample Code 5-7

Proprietary content removed.

Proprietary code removed.

Code Sample 5-7. Creating and releasing a message queue object

Sample Code 5-8

Proprietary content removed.

Proprietary code removed.

Code Sample 5-8. Defining messages in a group

Sample Code 5-9

Proprietary content removed.

Proprietary code removed.

Code Sample 5-9. Sending system messages

Sample Code 5-10

Proprietary content removed.

Proprietary code removed.

Code Sample 5-10. Receiving system messages

Section D Event Processing

Introduction Event processing is provided by the Event Observer Interface (EOI) API system. The EOI allows publishers to broadcast events to multiple subscribers. This section describes how the EOI API manages these events.

In this Section The following table lists the topics in this section.

Topic	Page
Event Observer Interface	151
Publisher and Subscriber Rules	152
Defining event codes	153
Subscribing to an event	154
Publishing an event	155
Acknowledging an event	156
Canceling an event	157
Canceling a subscription	158

Event Observer Interface

Introduction *Proprietary content removed.*

Implementation *Proprietary content removed.*

Publisher and Subscriber Rules

Introduction *Proprietary content removed.*

**Publisher
Rules** *Proprietary content removed.*

**Subscriber
Rules** *Proprietary content removed.*

Defining event codes

Introduction *Proprietary content removed.*

Sample Code *Proprietary content removed.*

Proprietary code removed.

Code Sample 5-11. Defining event codes

Subscribing to an event

Introduction *Proprietary content removed.*

Sample Code *Proprietary content removed.*

Proprietary code removed.

Code Sample 5-12. Subscribing to an event

Publishing an event

Introduction *Proprietary content removed.*

Sample Code *Proprietary content removed.*

Proprietary code removed.

Code Sample 5-13. Publishing an event

Acknowledging an event

Introduction *Proprietary content removed.*

Sample Code *Proprietary content removed.*

Proprietary code removed.

Code Sample 5-14. Acknowledging an event

Canceling an event

Introduction *Proprietary content removed.*

Sample Code *Proprietary content removed.*

Proprietary code removed.

Code Sample 5-15. Canceling an event

Canceling a subscription

Introduction *Proprietary content removed.*

Sample Code *Proprietary content removed.*

Proprietary code removed.

Code Sample 5-16. Canceling a subscription

Section E RTC, Alarm, Timers, and Persistent Bits

Introduction The following API subsystems provide functionality to manage real-time clocks (RTCs), alarms, persistent bits, and timers:

Proprietary content removed.

This chapter describes how to use these API subsystems to manage RTCs, alarms, and persistent bits.

In this Section The following table lists the topics in this section.

Topic	Page
Persistent Bits and Registers	160
Hardware Timers	161
Configuring hardware timers	162
Accessing persistent registers	163
Accessing digital shadow registers	164
Setting the time, date, and alarm values for the user clock	165

Persistent Bits and Registers

Introduction *Proprietary content removed.*

Copy controller *Proprietary content removed.*
For sample code, see “Accessing digital shadow registers” on page 164.

Registers *Proprietary content removed.*
For detailed information about these hardware registers, refer to the *NEXUS-6 Product Data Sheet*.

Persistent bits *Proprietary content removed.*

RTC Seconds Register *Proprietary content removed.*

RTC Alarm Register *Proprietary content removed.*
For more information on the user clock alarm, see “Setting the time, date, and alarm values for the user clock” on page 165.

Hardware Timers

Proprietary content removed.

For more information, see “Configuring hardware timers” on page 162 and *Tyrell SDK API Reference*.

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Configuring hardware timers

Introduction *Proprietary content removed.*

Sample Code *Proprietary content removed.*

Proprietary code removed.

Code Sample 5-17. Acquiring and configuring timers

Accessing persistent registers

Introduction *Proprietary content removed.*

For detailed information about these enumerated types, refer to the *Tyrell SDK API Reference*.

Sample Code *Proprietary content removed.*

Proprietary code removed.

Code Sample 5-18. Accessing persistent registers

Accessing digital shadow registers

Introduction *Proprietary content removed.*

For more information about individual functions, refer to the Device Driver API Time module in the *Tyrell SDK API Reference*.

Writing *Proprietary content removed.*

Reading *Proprietary content removed.*

Sample Code *Proprietary content removed.*

Proprietary code removed.

Code Sample 5-19. Accessing digital shadow registers

Limitations *Proprietary content removed.*

Setting the time, date, and alarm values for the user clock

Introduction *Proprietary content removed.*

Time and Date *Proprietary content removed.*

**Alarm
Configurations** *Proprietary content removed.*

Alarm Time *Proprietary content removed.*

Stopwatch *Proprietary content removed.*

Section F Power Management

Introduction The following API subsystems provide functionality to manage power state changes and control battery charging:

- ◆ Operating System (OS) Power Management Interface (PMI)
- ◆ DDI Battery Charger
- ◆ DDI Battery Charger Monitor (BCM)

This section describes these API subsystems.

In this Section The following table lists the topics in this section.

Topic	Page
Power Management Interface	167
Battery Charger and Battery Charger Monitor	168
Battery Charging FAQ	169
Enabling battery charging	170

See Also For information about battery charging hardware, refer to the *NEXUS-6 Product Data Sheet*.

Power Management Interface

Introduction *Proprietary content removed.*

Clients *Proprietary content removed.*

Subscribers *Proprietary content removed.*

For more information about the EOI, see “Event Processing” on page 150.

**PMI
Synchronous
API** *Proprietary content removed.*

**PMI
Asynchronous
API** *Proprietary content removed.*

Function Calls *Proprietary content removed.*

Clock Speed *Proprietary content removed.*

For more information, refer to the *NEXUS-6 Product Data Sheet*.

**Performance
Levels** *Proprietary content removed.*

Sample Code *Proprietary content removed.*

Proprietary code removed.

Code Sample 5-20. Sending a synchronous PMI client request

Proprietary content removed.

Proprietary code removed.

Code Sample 5-21. Sending an asynchronous PMI client request

Proprietary content removed.

Proprietary code removed.

Code Sample 5-22. Subscribing to a PMI event

Battery Charger and Battery Charger Monitor

Introduction *Proprietary content removed.*

Configuration Parameters *Proprietary content removed.*

For more information, see “Enabling battery charging” on page 170.

Battery Charger States *Proprietary content removed.*

Battery Charging FAQ

Proprietary content removed.

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Enabling battery charging

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

Sample Code *Proprietary content removed.*

Proprietary code removed.

Code Sample 5-23. Configuring battery charging

Section G Memory Management

Introduction The following API subsystems provide functionality to manage system memory:

- ◆ OS Dynamic Memory Interface
- ◆ OS Virtual Memory Interface

This section describes these API subsystems.

In this Section The following table lists the topics in this section.

Topic	Page
Dynamic Memory Interface (DMI)	172
Handle Heap	175
Memory Analysis	174
Handle Heap	175
Direct Memory Access	178
DMA and Data Cache Interaction	179

See Also For detailed information about the source code described in this section, refer to the *Tyrell SDK API Reference*.

Dynamic Memory Interface (DMI)

Introduction *Proprietary content removed.*

New Features *Proprietary content removed.*

SDRAM vs. NOSDRAM *Proprietary content removed.*

Initialization *Proprietary content removed.*

API Description *Proprietary content removed.*

How DMI Works *Proprietary content removed.*

Error Handling *Proprietary content removed.*

Virtual Memory Interface (VMI)

Introduction *Proprietary content removed.*

Memory Sections *Proprietary content removed.*

Initialization *Proprietary content removed.*

API Description *Proprietary content removed.*

Memory Placement *Proprietary content removed.*

How VMI Works *Proprietary content removed.*

Paging Area *Proprietary content removed.*

Memory Analysis

Static Memory Analysis *Proprietary content removed.*

Reducing Static Memory Usage *Proprietary content removed.*

Dynamic Memory Analysis *Proprietary content removed.*

Page Faults per Second *Proprietary content removed.*

Example *Proprietary content removed.*

Handle Heap

Introduction *Proprietary content removed.*

Concepts *Proprietary content removed.*

Organization *Proprietary content removed.*

Caching *Proprietary content removed.*

**Handle Heap
Size** *Proprietary content removed.*

Determining the number of handles

Introduction *Proprietary content removed.*

Font Caches *Proprietary content removed.*

Bitmap Cache *Proprietary content removed.*

**UIM Widgets
and Classes** *Proprietary content removed.*

**Handle Utility
Calls** *Proprietary content removed.*

Summary *Proprietary content removed.*

Determining the size of the handle heap

Introduction *Proprietary content removed.*

**Font Caches
and Glyph
Bitmaps** *Proprietary content removed.*

**Bitmap Cache
Size** *Proprietary content removed.*
For more information about determining the size of a bitmap, see “Bitmap
Drawing” on page 78.

**UIM Widgets
and Classes** *Proprietary content removed.*

**Handle Utility
Calls** *Proprietary content removed.*

Summary *Proprietary content removed.*

Direct Memory Access

Introduction *Proprietary content removed.*

Command Chains *Proprietary content removed.*

Hardware *Proprietary content removed.*

Controller Commands *Proprietary content removed.*

For more information about these command bits, refer to the *NEXUS-6 Product Data Sheet*.

Example *Proprietary content removed.*

For more information about this API, refer to the *Tyrell SDK API Reference*.

Limitations *Proprietary content removed.*

DMA and Data Cache Interaction

Introduction *Proprietary content removed.*

dCache *Proprietary content removed.*

Write Buffer *Proprietary content removed.*

Setup *Proprietary content removed.*

**Cache and
Write-Buffer
Control** *Proprietary content removed.*

Section H Headphone Short Detection

Introduction The following API subsystems provide functionality to control the headphone short detect driver:

Proprietary content removed.

This section describes how to use these API systems to control the headphone short detect driver.

In this Section The following table lists the topics in this section.

Topic	Page
Overview	181
Enabling headphone short detect	182
Configuring headphone short detect	183
Testing headphone short detect	184

See Also For detailed information about the source code described in this section, refer to the *Tyrell SDK API Reference*.

Overview

Introduction *Proprietary content removed.*

**Short
Handling** *Proprietary content removed.*

Limitations *Proprietary content removed.*

Enabling headphone short detect

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

Sample Code *Proprietary content removed.*

Proprietary code removed.

Code Sample 5-24. Enabling common mode amplifier short detection

Proprietary content removed.

Proprietary code removed.

Code Sample 5-25. Disabling common mode amplifier short detection

Configuring headphone short detect

Introduction *Proprietary content removed.*

Sample Code *Proprietary content removed.*

Proprietary code removed.

Code Sample 5-26. Configuring short detection levels

Testing headphone short detect

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

CHAPTER 6

Third Party Applications

Introduction This chapter provides the information necessary to manage portable device systems using the Tyrell SDK.

In this Chapter The following table lists the sections in this chapter.

Section	Topic	Page
A	SRS Labs	186
B	Audible	187
C	Rhapsody Optimized PlaysForSure	188

See Also

- ♦ For detailed information about the APIs described in this chapter, refer to the *Tyrell SDK API Reference*.
- ♦ For information about the *Tyrell SDK API Reference*, see “SDK API Reference” on page 30.

Section A SRS Labs

Proprietary content removed.

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Section B Audible

Introduction *Proprietary content removed.*

**Audible
Feature
Summary** *Proprietary content removed.*

**Adding
Audible to
Your Custom
Player** *Proprietary content removed.*

**Audible
Activation
Record** *Proprietary content removed.*

**Audible
Position File** *Proprietary content removed.*

**Audible
Hostlink
Modifications** *Proprietary content removed.*

**Audible Player
Modifications** *Proprietary content removed.*

Section C Rhapsody Optimized PlaysForSure

**SDK
Supported
Features** *Proprietary content removed.*

**Configurable
Code** *Proprietary content removed.*
See doxygen output for API information.

**Customer
Work Items** *Proprietary content removed.*

Not Supported *Proprietary content removed.*

APPENDIX A

Common Procedures

Introduction This appendix describes procedures that are common to the Tyrell SDK.

In this Appendix The following table lists the sections in this appendix.

Section	Topic	Page
A	Downloading binaries	190
B	Entering recovery mode	191
C	Windows Media DRM 10 Procedures	192
D	Compliance Testing	198

Section A Downloading binaries

- Introduction** The Tyrell SDK example host application is used to download binaries to the development system. This section describes how to download the binaries.
- Before You Begin** Verify the NEXUS-6 Development System is configured according to the default configurations described in Appendix B, “Development System Default Configurations.”
- Procedure** *Proprietary content removed.*

Section B Entering recovery mode

Introduction *Proprietary content removed.*

Procedure *Proprietary content removed.*

Issues *Proprietary content removed.*

Resolution Verify your development system is configured as described in Appendix B, “Development System Default Configurations” and then try again to enter recovery mode.

Section C Windows Media DRM 10 Procedures

Introduction This section provides the procedures necessary to obtain from Microsoft Windows Media DRM 10 for Portable Devices and to store the Windows Device Authorization Certificate (DAC) within your custom project.

In this Section The following table lists the topics in this section.

Topic	Page
Obtaining Windows Media DRM 10 for Portable Devices	193
Obtaining a DACResponse file	194
Creating an unsigned Device Certificate Template	195
Creating a signed Device Certificate Template	196
Creating and storing the build resource files	197

Obtaining Windows Media DRM 10 for Portable Devices

Introduction To distribute WMDRM 10 on devices that can store and play protected content, you must obtain a WMDRM 10 device certificate for Portable Devices.

Procedure To obtain WMDRM 10 for Portable Devices:

Step 1 Request from Microsoft a Windows Media DRM 10 for Devices Development and Interim Product Distribution Agreement by completing the License Request Form located at the following URL: www.microsoft.com

Once Microsoft approves the license agreement, you receive an email from *wmla@microsoft.com* that includes instructions on how to download additional materials from the Microsoft SpecServer, as well as instructions for requesting a device certificate.

! **IMPORTANT** This process may take several weeks to complete. We encourage you to submit your request as soon as possible. You will not be able to use online music services until you obtain a device certificate and complete the procedures described in this section.

Step 2 Access the Microsoft SpecServer using the instructions provide to you by Microsoft.

Step 3 Download the WMDRM 10 OEM installer package.

Step 4 Extract the files to your computer by double-clicking the installer package.

The WMDRM 10 OEM Kit is extracted to the specified location on your computer.

✍ **NOTE** If you used the default settings, the WMDRM 10 OEM Kit is extracted into the WMDRMPD_OEM_Kit directory. The WMDRMPD_OEM_Kit directory name is used within this document to refer to the WMDRM 10 OEM Kit directory.

Obtaining a DACResponse file

Introduction The DACResponse file is an XML file that is used to sign the Device Certificate Template (DCT) described in “Creating an unsigned Device Certificate Template” on page 195. You must obtain the DACResponse file from Microsoft.

 **NOTE** To receive and install a Windows Device Authorization Certificate (DAC), you must execute the Microsoft Windows Media DRM 10 for Devices Final Product Distribution Agreement, which is included with WMDRM 10 for Portable Devices, or be named as a *Third-Party Installer* or as an *Authorized Contractor* under another company’s WMDRM 10 Final Distribution Agreement.

Procedure To obtain a DACResponse file:

Step 1 Complete the Certificate Request Form that is provided with the WMDRM 10 OEM kit.

Step 2 Generate the PrivateKeyFile and the DACRequestFile files.

- A. Click the Windows **Start** button, and select **Run**.
- B. Type **cmd**, and click **OK**.
- C. Change to the WMDRMPD_OEM_Kit\Tools directory, and execute the following command:

```
generatedacrequest.exe -r:DACRequestFile -p:PrivateKeyFile
```

The DACRequestFile and PrivateKeyFile files are created in the WMDRMPD_OEM_Kit\Tools directory. For more information about the GenerateDACRequest tool, refer to the WMDRM_PD.chm file located in the WMDRMPD_OEM_Kit\Documents directory.

Step 3 Email the completed Certificate Request Form and the DACRequestFile to wmla@microsoft.com.

 **IMPORTANT** Do not submit PrivateKeyFile to Microsoft. This file should remain proprietary to your company.

Once the Certificate Request Form and DACRequestFile are processed, Microsoft emails to you a DACResponse file (DACResponse.xml).

Step 4 Place the DACResponse.xml file in the WMDRMPD_OEM_Kit\Tools directory, and proceed to “Creating a signed Device Certificate Template” on page 196.

For more information about this procedure, refer to AcquiringDACResponse.doc located in the WMDRMPD_OEM_Kit\Documents directory.

Creating an unsigned Device Certificate Template

Introduction The Device Certificate Template (DCT) is an XML document that outlines information about your device model and feature set. After the DCT is created, it must be signed by Microsoft using the DACResponse file as described in “Creating a signed Device Certificate Template” on page 196.

Procedure To create an unsigned DCT:

Step 1 Copy the following text from AcquiringDACResponse.doc (located in the WMDRMPD_OEM_Kit\Documents directory) and paste it into a text file.

```
<UNSIGNEDTEMPLATE>
...
</UNSIGNEDTEMPLATE>
```

Step 2 Modify the values of each tag as necessary. Microsoft recommends that you include all information listed; however, you may include only the following tags if desired:

- ◆ <SECURITYLEVEL>
- ◆ <FEATURES>
- ◆ <LIMITS>

! **IMPORTANT** You must change the <SECURITYLEVEL> tag to include the security level provided by Microsoft in the DACResponse file; for example: <SECURITYLEVEL>2000</SECURITYLEVEL>

Step 3 Save the file to the WMDRMPD_OEM_Kit\Tools directory using the following file name: UnsignedTemplate

 **NOTE** This file must be saved using Unicode encoding. To save the file using Unicode encoding, use an application such as Notepad, select **Save As** from the File menu, and then select **Unicode** from the Encoding pull-down menu. In addition, you must remove all tabs and spaces between tags.

For more information about the contents of an unsigned device certificate template and valid values for each tag, refer to the following files located in the WMDRMPD_OEM_Kit\Documents directory:

- ◆ WMDRM_PD.chm
- ◆ AcquiringDACResponse.doc

Creating a signed Device Certificate Template

Introduction Once you create the unsigned DCT, you must sign the DCT using the GenerateDevCertTemplate tool. The GenerateDevCertTemplate tool adds a group certificate to the unsigned template and then signs the template using the PrivateKeyFile. In addition to the signed DCT, DevCertTemplateFile, the GenerateDevCertTemplate tool creates a group private key, OemGroupKey.bin.

! **IMPORTANT** It is critical that you do not disclose private keys to individuals outside your organization.

For more information about the GenerateDevCertTemplate tool, refer to the WMDRM_PD.chm file located in the WMDRMPD_OEM_Kit\Documents directory.

Required Files The following files are used in this procedure:

- ◆ UnsignedTemplate
- ◆ DACResponse.xml
- ◆ PrivateKeyFile

 **NOTE** If you followed the procedures described in this document, these files are located in the WMDRMPD_OEM_Kit\Tools directory.

Procedure To create a signed DCT:

Step 1 Click the Windows **Start** button, and select **Run**.

Step 2 Type **cmd**, and click **OK**.

Step 3 Change to the WMDRMPD_OEM_Kit\Tools directory, and execute the following command:

```
GenerateDevCertTemplate.exe -d:DACResponseFile -u:UnsignedTemplate  
-f:PrivateKeyFile -t:DevCertTemplateFile -k:OemGroupKey.bin
```

The following files are created in the WMDRMPD_OEM_Kit\Tools directory:

- ◆ DevCertTemplateFile
- ◆ OemGroupKey.bin

For more information about these files, refer to AcquiringDACResponse.doc located in the WMDRMPD_OEM_Kit\Documents directory.

Creating and storing the build resource files

Introduction *Proprietary content removed.*

Required Files *Proprietary content removed.*

Procedure *Proprietary content removed.*

Section D Compliance Testing

Introduction *Proprietary content removed.*

**USB
Commands** *Proprietary content removed.*

**PlaysForSure
Requirements** *Proprietary content removed.*

**PlaysForSure
Logo** *Proprietary content removed.*

**PlaysForSure
Testing** *Proprietary content removed.*

**PlaysForSure
Testing with
the SCTL SDK** *Proprietary content removed.*

**PlaysForSure
Test Kit Hints** *Proprietary content removed.*

**Troubleshoot
PlaysForSure** *Proprietary content removed.*

APPENDIX B

Development System Default Configurations

Introduction This appendix describes the default configurations for the Replicant Engineering Mainboard and the different versions of the NEXUS-6 CPU Daughtercard. These default configurations are applied before the NEXUS-6 Development System is shipped.

In this Appendix The following table lists the sections in this appendix.

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A	Replicant Engineering Mainboard	200
B	NEXUS-169 CPU Daughtercard	201
C	NEXUS-100 CPU Daughtercard	202
D	Additional Add-On Daughtercards	203

Section A Replicant Engineering Mainboard

Introduction *Proprietary content removed.*

**Default
Configuration** *Proprietary content removed.*

Section B NEXUS-169 CPU Daughtercard

Introduction *Proprietary content removed.*

**Default
Configuration** *Proprietary content removed.*

Section C NEXUS-100 CPU Daughtercard

Introduction *Proprietary content removed.*

**Default
Configuration** *Proprietary content removed.*

Section D Additional Add-On Daughtercards

Introduction *Proprietary content removed.*

Parallel-to-Serial JTAG Board *Proprietary content removed.*

SDRAM Daughtercard *Proprietary content removed.*

NAND Flash Daughtercard *Proprietary content removed.*

STFM1000 Daughtercard *Proprietary content removed.*

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Glossary

AAC

An abbreviation for Advanced Audio Compression.

AC

An abbreviation for *alternating current*.

ADC

An abbreviation for analog-to-digital converter. Converts an analog signal to digital. See also **DAC**.

ADPCM

An abbreviation for adaptive differential pulse code modulation. See also **PCM**.

AHB

An abbreviation for AMBA High-performance Bus.

ASCII

An abbreviation for American Standard Code for Information Interchange.

ASI

An abbreviation for Audio Stream Interface. A Tyrell Component API that provides functionality to support streaming audio.

ASM

An abbreviation for Audio Stream Manager. A Tyrell Audio Stream Interface Component API. See also **ASI**.

ATA

An abbreviation for Advanced Technology Attachment.

BCM

An abbreviation for Battery Charge Monitor. A Tyrell Device Driver API.

BGA

An abbreviation for ball grid array.

capless

Capless is also known as *direct drive* or *DC-coupled*.

CMI

An abbreviation for Content Management Interface. A Tyrell Component API that provides functionality to support player content, such as media files.

CMOS

An abbreviation for complementary metal oxide semiconductor.

code page

Within the Tyrell documentation, a text file containing mapping information from DBCS to unicode.

codec

Short for coder/decoder.

CompactFlash

A digital camera memory card.

DAC

An abbreviation for digital-to-analog converter. Converts digital code to an analog signal.

DAI

An abbreviation for Database Access Interface. A Tyrell Content Management Interface API.

daughtercard

A printed circuit board that plugs into another printed circuit board. See also **PCB**.

DBCS

An abbreviation for Double Byte Character Set.

DC

An abbreviation for direct current. Also known as continuous current.

DC/DC converter

A circuit that converts direct current from one voltage to a another.

dCache

Short for data cache.

DFFI

An abbreviation for Decoder File Format Interface. A Tyrell Audio Stream Interface Component API.

DMA

An abbreviation for direct memory access.

DMI

An abbreviation for Dynamic Memory Interface. A Tyrell Operating System API that provides memory allocation.

DPC

An abbreviation for Deferred Procedure Call. A Tyrell Operating System API for deferring low-priority functions to a set of pre-defined system tasks.

DRAM

An abbreviation for dynamic RAM.

DRI

An abbreviation for Digital Radio Interface. A Tyrell Device Driver API.

DRM

An abbreviation for Windows Media *digital rights management*. A technology used to protect and securely deliver media for playback on an audio device.

DSP

An abbreviation for digital signal processing.

DSR

An abbreviation for deferred service routine. A routine invoked in response to a request by an ISR. See also **IRQ** and **ISR**.

ECC

An abbreviation for error-correcting code.

EEPROM

An abbreviation for electrically-erasable programmable read-only memory.

EFFI

An abbreviation for Encoder File Format Interface. A Tyrell Audio Stream Interface Component API.

EMI

An abbreviation for external memory interface.

EOI

An abbreviation for Event Observer Interface. A Tyrell Operating System API.

ETM

An abbreviation for embedded trace macrocell.

FFI

An abbreviation for file format interface.

FM

An abbreviation for frequency modulation.

font

Within the Tyrell documentation, a collection of images (or glyphs) used to visually represent character data.

GEQ

An abbreviation for graphics equalizer.

gfx

Short for graphics and is used to identify Tyrell's Graphics API.

glyph

Within the Tyrell documentation, an image that is part of a font.

GPIO

An abbreviation for General Purpose Input/Output.

GPMI

An abbreviation for general purpose media interface.

handle heap

Provides a mechanism to allocate memory from a heap.

HCM

An abbreviation for Hardware Condition Monitor. A Tyrell Operating System API.

host mode

The mode in which a device, such as the Tyrell development platform or an MP3 player, is connected to a host computer. In host mode, content can be

downloaded onto the device from the host computer. See also **player mode**, **MSC mode**, and **MTP mode**.

I²C

An abbreviation for Inter-IC or I²C-bus. A standard serial bus used for embedded applications.

iCache

Short for instruction cache.

ICOLL

An abbreviation for interrupt collector.

ID3 tag

Identifies information added to the end of an MP3 file.

IDE

An abbreviation for integrated development environment.

IRQ

An abbreviation for interrupt requests. The technology used by hardware components to request computing time from the CPU. See also **DSR** and **ISR**.

ISR

An abbreviation for interrupt service routine. A routine invoked in response to an IRQ. Also known as an interrupt handler. See also **IRQ** and **DSR**.

JPEG

An abbreviation for Joint Photographic Experts Group. A standard for compressing photographic images.

LCD

An abbreviation for liquid crystal display.

LCDIF

An abbreviation for LCD interface. See also **LCD**.

LDL

An abbreviation for Logical Device Layer. A Tyrell Operating System API.

LED

An abbreviation for light emitting diode.

LFN

An abbreviation for long file name.

Li-ion

An abbreviation for lithium ion. A type of a battery composed of lithium.

LRADC

An abbreviation for low-resolution ADC. See also **ADC**.

m3u

The file extension for an MP3 playlist.

MBCS

An abbreviation for Multiple Byte Character Set.

MCU

An abbreviation for microcontroller unit.

MIPS

An abbreviation for million instructions per second.

MLC

An abbreviation for multilevel cell. A technology that expands memory cell capacity.

MMC

An abbreviation for MultiMedia card. An external flash memory storage device. See also **SD**.

MMU

An abbreviation for memory management unit. A hardware component that manages virtual memory.

MP3

An abbreviation for MPEG Layer 3. An audio file format.

MS

An abbreviation for Memory Stick.

MSC

An abbreviation for mass storage class. A protocol for transferring data.

MSC mode

The mode in which a device, such as the Tyrell development platform or an MP3 player, operates using the MSC technology. See also **host mode**, **player mode**, and **MTP mode**.

MTP

An abbreviation for Media Transfer Protocol. A Microsoft protocol for transferring media.

MTP mode

The mode in which a device, such as the Tyrell development platform or an MP3 player, operates using the MTP technology. See also **player mode**, **MSC mode**, and **host mode**.

MVA

An abbreviation for modified virtual address.

NAND

Short for not AND. An electronic logic gate.

NTC

An abbreviation for NTC. A type of thermistor. See **thermistor**.

NiMH

An abbreviation for nickel-metal hydride. A material used in batteries.

NOR

Short for not OR. An electronic logic gate.

ODM
An abbreviation for original design manufacturer.

OEM

An abbreviation for original equipment manufacturer.

OTG

An abbreviation for On-The-Go. Allows portable devices to communicate directly without a PC.

PCB

An abbreviation for printed circuit board.

PCM

An abbreviation for pulse code modulation.

PD

An abbreviation for portable device.

PDA

An abbreviation for personal digital assistant.

PDDRM

Short for Microsoft's DRM for portable devices technology.

PHY

An abbreviation for physical layer.

PID

An abbreviation for Proportional-Integral-Derivative. Measures the input and controls the output of a process with the goal of maintaining a target value.

player mode

The mode in which a device, such as the Tyrell development platform or an MP3 player, is used as a media player. See also **host mode**, **MSC mode**, and **MTP mode**.

PLL

An abbreviation for phase-locked loop. An electronic circuit that maintains a constant lock on an input frequency.

PMI

An abbreviation for Power Management Interface. A Tyrell Operating System API.

PQI

An abbreviation for Play Queue Interface. A Tyrell Component API for managing play queues.

PPM

An abbreviation for Post-Processing Manager. A Tyrell Component API.

pswitch

An abbreviation for process switch.

PTC

An abbreviation for positive temperature coefficient. A type of thermistor.

See also **thermistor**.

PWM

An abbreviation for pulse width modulation.

QFE

An abbreviation for Quick-Fix Engineering. A software update that resolves a specific engineering issue. Also known as Quality Fix Engineering.

QFP

An abbreviation for Quad Flat Package. See also **TQFP**.

QMI

An abbreviation for Query Manager Interface. A Tyrell Content Management Interface API.

RAM

An abbreviation for random access memory. A type of memory that holds processes and data for programs that are currently running.

RDS

An abbreviation for Radio Data System. A Tyrell Component API.

ROM

An abbreviation for read-only memory.

RTC

An abbreviation for real-time clock. A computer clock that tracks the current time even when the computer is turned off.

RTOS

An abbreviation for real-time operating system.

S/PDIF

An abbreviation for Sony/Philips Digital Interface. A standard file format for audio transfer.

SD

An abbreviation for Secure Digital. A flash memory storage device.

SDK

An abbreviation for software development kit.

SDRAM

An abbreviation for synchronous dynamic RAM.

SMI

An abbreviation for Sort Module Interface. A Tyrell Content Management Interface API.

SmartMedia card

External memory card comprised of a NAND flash EEPROM chip encased in a thin plastic card. See also **SSFDC**.

SMB

An abbreviation for System Management Bus.

SNR

An abbreviation for signal-to-noise ratio. The amount of noise added by an electrical circuit that reproduces an audio signal.

SoC

An abbreviation for system-on-chip.

SPI

An abbreviation for serial peripheral interface.

SRAM

An abbreviation for static RAM. A type of **RAM** that holds data as long as power is supplied to the circuit. See also **DRAM**.

SRC

An abbreviation for sample rate converter. The mechanism by which the sample rate of a data stream is converted from one rate to another.

SSFDC

An abbreviation for Solid State Floppy Disk Card.
See also **SmartMedia card**.

SSP

An abbreviation for synchronous serial port.

subclassing

A subtyping technique used in object-oriented programming.

thermistor

A type of resistor that measures temperature changes.

THD

An abbreviation for total harmonic distortion.

THI

An abbreviation for ThreadX Interface. A Tyrell Operating System API that provides an interface to the ThreadX RTOS. See also **RTOS**.

TQFP

An abbreviation for thin Quad Flat Package. See also **QFP**.

TSS

An abbreviation for Telemetry Subsystem. A Tyrell Component API.

UART

An abbreviation for universal asynchronous receiver-transmitter. Computer hardware that translates between parallel bits of data and serial bits.

USB

An abbreviation for Universal Serial Bus.

UTMI

An abbreviation for USB 2.0 transceiver macrocell interface.

VFIR

An abbreviation for Very Fast Infrared.

VMI

An abbreviation for Virtual Memory Interface. A Tyrell low-level API for the allocation of large blocks of memory.

widget

User interface components with which the user interacts.

WMA

An abbreviation for Windows Media Audio.

WMDM

An abbreviation for Windows Media Device Manager.

WMDRM

An abbreviation for Windows Media DRM. See also **DRM**.

wpl

File extension for Windows Media Player playlists.

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